PG No. 12　D_4　422　[tetragonal]

* character table

D_4	$1(1)$	$2_{001}(1)$	$2_{100}(2)$	$2_{110}(2)$	$4^+_{001}(2)$
A_1	1	1	1	1	1
A_2	1	1	−1	−1	1
B_1	1	1	1	−1	−1
B_2	1	1	−1	1	−1
E	2	−2	0	0	0

* polar ↔ axial conversion

A_1 (A_1)　B_1 (B_1)　E (E)　A_2 (A_2)　B_2 (B_2)

* symmetric product

	A_1	A_2	B_1	B_2	E
A_1	A_1	A_2	B_1	B_2	E
A_2		A_1	B_2	B_1	E
B_1			A_1	A_2	E
B_2				A_1	E
E					$A_1 + B_1 + B_2$

* anti-symmetric product

A_1	A_2	B_1	B_2	E
–	–	–	–	A_2